

(Reaffirmed 2014)

**IS 302-2-204 : 1994**

भारतीय मानक

( Reaffirmed 2004 )

(Reaffirmed 2019)

घरेलू और समान विद्युत साधनों की सुरक्षा

भाग 2 विशेष अपेक्षाएँ

अनुभाग 204 पानी के लिए विद्युत बायलर

*Indian Standard*

## SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

**PART 2 PARTICULAR REQUIREMENTS**

**Section 204 Electric Water Boilers**

UDC 641.542.1 - 83 : 614.825

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**BUREAU OF INDIAN STANDARDS**  
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NEW DELHI 110002

March 1994

Price Group 4

Electrical Appliances Sectional Committee, ETD 32

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrotechnical Division Council.

This standard covers the general and safety requirements for Electric Water Boilers to ensure personal safety against electric shock, effect of excessive temperature and fire, etc.

For the sake of convenience of reference the format of the standard has been followed with the pattern followed in IS 302-1 ( 1979 ) 'Safety of household and similar electrical appliances : Part 1 General requirements ( *fifth revision* )'. The performance requirements and the detail list of performance characteristics along with the methods for evaluation of these characteristics are covered in a separate standard IS 3412 : 1993 'Specification for electric water boilers ( *second revision* )'.

This standard is to be read in conjunction with IS 302-1 ( 1979 ). For the sake of convenience, where the clause of this standard corresponds to that of IS 302-1 ( 1979 ), instead of reproducing full text of each clause of IS 302-1 ( 1979 ) which are applicable ( which means that relevant provision of the clause applies ) or not applicable and the sub-clauses or portion thereof which are not applicable are indicated as under:

- a) In case of a clause where it is applicable or not applicable, the wording used is 'This clause of IS 302-1 ( 1979 ) is applicable/not applicable'.
- b) In case of a sub-clause or part thereof 'Not applicable'.

Wherever a sub-clause of IS 302-1 ( 1979 ) is to be replaced by a new text, it has been indicated as under:

Replacement — Followed by the new text.

Any addition to the existing provision of sub-clause of IS 302-1 ( 1979 ) has been indicated as under:

Addition — Followed by the text of additional clause.

Clauses/Tables which are additional to those of IS 302-1 ( 1979 ) are numbered starting from 101 and additional sub-clauses are numbered with the main clause number followed by 101, 102, 103, etc, for example 2.101 should however any deviations exist between IS 302-1 ( 1979 ) and this standard, the provision of latter shall apply.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values ( *revised* )'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

IS 302-2-204 : 1994

# *Indian Standard*

## SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

### PART 2 PARTICULAR REQUIREMENTS

#### Section 204 Electric Water Boilers

#### 1 SCOPE

##### 1.1 Replacement

This standard covers the general and safety requirements for electric water boilers of rated loading not exceeding 5.0 kW and connection to supplies of voltages not exceeding 250 V ac single phase 50 Hz or dc.

##### 1.2 Replacement

This standard does not apply to :

- Appliances intended to be used in locations where special conditions prevail, such as the presence of a corrosive or explosive atmosphere (dust, vapour or gas);
- Appliances for high-frequency heating;
- Appliances of the electrode type;
- Storage water heaters, which are covered by a separate standard;
- Instantaneous water heaters, which are covered by separate standard;
- With rated capacity not exceeding 10 l are covered under a separate Indian Standard.

#### 2 TERMINOLOGY

This clause of IS 302-1(1979) is applicable except as follows :

2.9, 2.11, 2.25 and 2.28 not applicable.

##### 2.29 Replacement

Conditions of Adequate Heat Discharge — Water boilers are operated filled with a quantity of initially cold water equal to its rated capacity and with the lid, where provided, closed.

2.30, 2.31, 2.33, 2.34 and 2.46 not applicable.

##### Additional Sub-clauses

**2.101 Electric Water Boiler** — An electrically heated vessel with free outlet for the steam, intended for boiling water for bathing, washing fabrics, preparing beverages, etc.

**2.102 Clamp-on Heating Element** — A heating element clamped or similarly attached to the outer surface of the water container.

**2.103 Immersion Heating Element** — A totally enclosed heating element the heating portion of which is immersed in water.

**2.104 Rated Water Capacity** — The quantity of water which the water boiler is designed to hold and which is marked on the rating plate.

**2.105 Fixed Water Boiler** — It denotes an appliance for permanent connection to the water supply mains, which if filled through a valve in the inlet pipe and is open to the atmosphere, so that under no conditions of use can the pressure at the surface of the water be other than atmospheric.

#### 3 GENERAL REQUIREMENTS

This clause of IS 302-1(1979) is applicable except as follows :

##### 3.1 Para 3 Replacement

In general, compliance is checked by carrying out the relevant tests specified.

#### 4 GENERAL NOTES ON TESTS

This clause of IS 302-1(1979) is applicable except as follows :

4.2 Note 3 and Note 4 not applicable.

##### Addition

For water boilers intended to be immersed in water for cleaning, two additional samples are required for tests of sub-clause 15.101.

4.8, 4.10 and 4.13 Not applicable.

##### Additional Sub-clause

**4.101** The appliances are tested as portable appliances, unless it is clear from their design and the manufacturer's instructions that they are intended to be fixed to a supporting surface.

#### 5 RATING

This clause of IS 302-1(1979) is applicable except as follows.

##### 5.1 Replacement

The rated voltage shall not exceed 250 V. The preferred rated voltage is 240 V.

**IS 302-2-204 : 1994**

**5.2 Rated Input**

The rated input of water boilers shall not exceed 5.0 kW.

**5.2.1** Recommended input for various capacities of water boilers ( *see 5.101* ) are as follows :

| Rated Capacity<br>litres | Recommended Input<br>kilowatt |
|--------------------------|-------------------------------|
| 15                       | 1.5                           |
| 20                       | 2.0                           |
| 25                       | 2.5                           |
| 30                       | 2.5                           |
| 35                       | 3.0                           |
| 40                       | 3.0                           |
| 50                       | 3.0                           |

**5.2.2** and **5.2.3** not applicable

**Additional Sub-clause**

**5.101 Rated Capacity**

The rated capacity for water boilers shall be one of the following :

15, 20, 25, 30, 35, 40 and 50 litres.

**6 CLASSIFICATION**

This clause of IS 302-1(1979) is applicable except as follows :

**6.1** (a) (2) and (3), and **6.1** (b) (4) Not applicable.

**Addition**

**NOTE** — This standard recognises appliances which are intended to be cleaned by pouring water in the inner container but it does not recognize water tight appliances.

**7 MARKING**

This clause of IS 302-1(1979) is applicable except as follows.

**7.1 Replacement**

The water boilers shall be marked with the following information:

- a) Rated voltage in volts,
- b) Rated input in watts or kilowatts,
- c) Rated capacity,
- d) Name of the manufacturer or his trade-mark,
- e) Maker's model or type reference, and
- f) Country of origin.

**Addition**

Appliances intended to be used with a connector incorporating a thermostat shall be marked with the substance of the following warning, unless this warning is given in the instruction sheet.

This appliance must only be used with the appropriate connector.

**7.3, 7.4** and **7.8** Not applicable.

**7.12 Addition**

The instruction sheet should state that the appliance should be cleaned by pouring water in the inner container.

**Additional Sub-clause**

**7.101** The water boilers may also be marked with the Standard Mark.

**8 PROTECTION AGAINST ELECTRIC SHOCK**

This clause of IS 302-1(1979) is applicable except as follows :

**8.2, 8.3** and **8.8** not applicable.

**9 STARTING OF MOTOR-OPERATED APPLIANCE**

This clause of IS 302-1(1979) is not applicable.

**10 INPUT AND CURRENT**

This clause of IS 302-1(1979) is applicable except as follows :

**10.2** Not applicable.

**11 TEMPERATURE RISE**

This clause of IS 302-1(1979) is applicable except as follows :

**11.2** Para 3 (a) Replacement

Water boilers provided with a temperature limiter and normally used on a floor or a table are placed on the floor of the test corner at a distance of about 15 cm from the walls of the test corner.

**NOTE** — This distance includes the natural projection such as hooks, bolts, appliance inlets, etc, of the appliances.

Para 4 Not applicable.

**11.3** Para 6 Not applicable.

**11.5** and **11.6** Not applicable.

**11.7 Replacement**

Water boiler shall be tested for 15 minutes after the water in the container has attained a temperature of 95°C or if this temperature cannot be attained, after water has attained its maximum temperature.

**11.9** and **11.10** Not applicable.

**12 OPERATION UNDER OVERLOAD CONDITIONS OF APPLIANCES WITH HEATING ELEMENT**

This clause of IS 302-1(1979) is applicable except as follows :

**IS 302-2-204 : 1994**

**12.2** Para 1 Not applicable.

**12.3** Not applicable.

**13 ELECTRICAL INSULATION AND LEAKAGE CURRENT AT OPERATING TEMPERATURE**

This clause of IS 302-1(1979) is applicable except as follows.

**13.1 Replacement**

Compliance is checked by the tests of **13.2** and **13.3**, the water boiler being operated under the conditions specified in **11**.

**14 RADIO AND TELEVISION INTERFERENCE SUPPRESSION**

**14.1** This clause of IS 302-1(1979) is applicable.

**15 MOISTURE RESISTANCE**

This clause of IS 302-1(1979) is applicable except as follows.

**15.2.2** *Not applicable*

**15.2.3 Replacement**

Test for Splashing

The water boiler shall be allowed to stand on a level surface and a volume of water equal to 95 percent of the rated capacity is poured in. No overflow of water shall take place.

The water boiler shall then be filled to 85 percent of the rated capacity, and with the lid on, if any, it is allowed to stand on a sheet of blotting paper of suitable dimensions and connected to a supply voltage adjusted to given rated input. The water boiler shall be maintained on circuit for a period of two minutes after boiling has commenced. There shall be no signs of moisture on the blotting paper after further period of 20 minutes.

**16 INSULATION RESISTANCE AND ELECTRIC STRENGTH**

This clause of IS 302-1(1979) is applicable except as follows.

**16.1** Para 2 Replacement

Compliance is checked by the tests of **16.2** and **16.4** which are made on the cold water boiler not connected to the supply, after the test given in **15.4** in the humidity cabinet or in the room in which the sample was brought to the prescribed temperature, after reassembly of those parts which may have been removed.

**16.3** Not applicable.

**17 OVERLOAD PROTECTION**

This clause of IS 302-1(1979) is applicable.

**18 . ENDURANCE** (Under consideration).

**19 ABNORMAL OPERATION**

This clause of IS 302-1(1979) is applicable except as follows.

**19.1 Replacement**

Appliances shall be so designed that the risk of fire mechanical damage impairing safety or the protection against electric shock as a result of abnormal or careless operation is obviated as far as is practicable.

Compliance is checked by the tests of **19.2** and, if necessary, by the test of **19.3**.

If, in any of the test, a non self resetting thermal cutout operates, an ejector type device operates, a heating element ruptures or if the current is otherwise interrupted before steady conditions are established, the heating period is considered to be ended, but if the interruption is due to the rupture of a heating element or of an intentionally weak part, the relevant test is repeated on a second sample which shall then also comply with the conditions specified in **19.11**.

Rupture of a heating element or of an intentionally weak part in the second sample will not in itself entail a rejection.

An intentionally weak part designed to fail under conditions of abnormal operation so as to prevent the occurrence of a condition which is unsafe within the meaning of this standard. Such as a resistor capacitor, thermal fuse or part of a component to be replaced, such as an inaccessible and non-resettable thermal cut-out incorporated in a motor.

Fuses, thermal cutouts, ejector type devices, overcurrent releases or the like, incorporated into the appliance, may be used to provide the necessary protection.

If more than one of the tests are applicable for the same appliance, these tests are made consecutively.

Provisions of **19.11** apply to all appliances.

**19.2** The water boilers are tested empty under conditions specified in **11**. The supply voltage is such that the input is 0.85 times the rated input if any, are in the open or closed position whichever imposes more severe conditions. The thermostat is short-circuited during the test.

If an ejector type device operates, or if the current is otherwise interrupted before steady conditions are established and is allowed to cool down to approximately room temperature, the appliance is then subjected to the test of **19.3**, **19.4** to **19.10** not applicable.

**20 STABILITY AND MECHANICAL HAZARDS**

This clause of IS 302-1(1979) is applicable except as follows.

**20.1** Para 5 Replacement

The water boiler shall be filled with the maximum amount of water it will hold in the tilted position. If

## IS 302-2-204 : 1994

castors or rollers are fitted they are rendered inoperative during the test.

**20.2** Not applicable.

## 21 MECHANICAL STRENGTH

This clause of IS 302-1(1979) is applicable except as follows :

### 21.1 Addition

The water boiler while standing on its feet or plinth without water, is dragged 10 times over a concrete floor backwards and forward for a distance of 1.5 metres, at a rate of approximately 15 metres per minute. After this test there shall be no damage to the water boiler nor loosening of the legs, castor or plinth fixings.

**NOTE** — The water boiler is dragged by the handle, rim or other suitable projection and not by the flexible cord.

Breakage of glass parts is neglected, provided that live parts do not become exposed so as to impair compliance with **8.1**, **15.1**, **15.2** and **15.101**.

## 22 CONSTRUCTION

This clause of IS 302-1(1979) is applicable except as follows.

### 22.9 Addition

Insulated fixed water boiler shall be so designed that condensed water can flow out, if otherwise there would be a possibility of such water affecting the electrical insulation. If a drain hole is provided for this purpose, it shall be at least 5 mm in diameter, or 20 sq mm in area with a width of at least 3 mm and so arranged that water can drain off without impairing the electrical insulation.

### Replacement

Compliance is checked by inspection and, if necessary, by measurement.

**22.12**, **22.15**, **22.20**, **22.24** and **22.25** Not applicable.

### 22.25 Replacement

Insulated or uninsulated fixed water boilers shall be so constructed that the container is always open to the atmosphere through an aperture at least 5 mm in diameter or 20 sq mm in area with a width of at least 3 mm. The aperture shall be so located that it is unlikely to be obstructed in normal use.

Compliance is checked by inspection and by measurement.

**22.29**, **22.31** and **22.32** Not applicable.

**22.101** The water boiler shall be of double-walled construction and provided with a lid. The lid shall have a knob or handle of heat insulating material.

**22.102** The design of the water boiler shall be such that when the lid, if placed in the intended position, emission of steam from the vents shall be directed away from the knob or handle provided on the lid.

**22.103** The lagging material, when provided, shall be packed and supported in a manner which precludes the possibility of large air pockets developing within it.

**22.104** All surfaces in contact with water shall be protected, if necessary, so that resistance to corrosion is not inferior to that of copper coated with a layer of 0.025 mm thickness of tin of 99.75 percent purity ( see IS 26 : 1979 ).

**22.105** For fixed water boilers having provision for discharging steam or overflowing water, the discharge orifice shall be at the base of the appliance and shall discharge vertically downwards.

**22.106** Fixed water boilers shall allow connection to the water supply mains in a reliable manner. In addition, they shall have reliable provisions for fixing them to a supporting surface.

Compliance with the requirements of **22.105** and **22.106** is checked by inspection.

**22.107** The water boiler shall be provided with a draw-off tap made of metal. The knob or handle of the tap shall be of heat insulating material. The tap shall be so located and designed that it shall lead the water clear off from the electrical components, wiring and fittings, and from the body of the container.

**22.108** Facility of Maintenance — Similar parts of water boilers of the same make, model and size shall be readily interchangeable. The parts which may require maintenance or replacement shall, as far as practicable, be so located as to ensure easy access.

**22.109** Inner containers shall be made of copper (conforming to IS 1550 : 1967) or non-magnetic stainless steel of Grade 04 Cr 18 Ni 11 or 07 Cr 18 Ni 19 conforming to IS 1570 (Part 5) : 1985.

The outer casing of the water boiler shall be made of corrosion resisting material or shall be adequately protected from corrosion on all surfaces. The lid shall be made of corrosion resisting material.

## 23 INTERNAL WIRING

This clause of IS 302-1(1979) is applicable except as follows.

**23.4** Not applicable.

## 24 COMPONENTS

This clause of IS 302-1(1979) is applicable except as follows :

**24.6**, **24.9** and **24.10** Not applicable.

### 24.101 Heating Units

The heating units shall be of immersion element type. It shall be so fixed as to be easily removable and replaceable.

## 25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CABLES AND CORDS

This clause of IS 302-1(1979) is applicable.

IS 302-2-204 : 1994

**25.6 Addition**

The supply cord external to the appliances (entry point into the appliance and the plug including any cord guard) shall be not less than 2.05 m in case the appliance is not provided with plug. In case the appliance is provided with a plug the length and the supply cord shall be not less than 2.0 m.

**26 TERMINALS FOR EXTERNAL CONDUCTORS**

This clause of IS 302-1(1979) is applicable.

**27 PROVISION FOR EARTHING**

This clause of IS 302-1(1979) is applicable.

**28 SCREWS AND CONNECTIONS**

This clause of IS 302-1(1979) is applicable.

**29 CREEPAGE DISTANCES AND CLEARANCES**

This clause of IS 302-1(1979) is applicable except as follows :

**29.1** Paras 2 and 3 Not applicable.

**30 RESISTANCE OF HEAT, FIRE AND TRACKING**

This clause of IS 302-1(1979) is applicable.

**31 RESISTANCE TO RUSTING**

This clause of IS 302-1(1979) is applicable.

**32 RADIATION HAZARDS**

This clause of IS 302-1(1979) is applicable.

**101 TESTS**

**101.0 Categories of Tests**

Tests are classified as type, acceptance and routine tests.

**101.1 Type Tests**

The tests specified in Table 101 shall constitute the type tests and shall be carried out on two samples of the same type and rating selected preferably at random from a regular production lot. Before commencement of the tests, the samples shall be visually examined and inspected for obvious visual defects in respect of components, parts and their assembly, construction, mechanical hazards, markings, provision, of suitable terminals for supply connections, earthing and the effectiveness of screws and connections. The external surface finish shall be even and free from finishing defects.

**101.1.1 Criteria of Acceptance**

Both samples shall successfully pass all the type tests for proving conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority, at its discretion, may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test(s) in which failure(s) had occurred. No failure shall be permitted in the repeat test(s).

**Table 101 Schedule of Type Test**

( Clause 101.1 )

| Sl No.                                                                       | Clause Reference |
|------------------------------------------------------------------------------|------------------|
| i) Protection against electric shock                                         | 8                |
| ii) Input                                                                    | 10               |
| iii) Temperature rise                                                        | 11               |
| iv) Operation under overload conditions                                      | 12               |
| v) Electrical insulation and leakage current at operating temperature        | 13               |
| vi) Radio and television interference suppression                            | 14               |
| vii) Moisture resistance                                                     | 15               |
| viii) Insulation resistance and electric strength (after humidity treatment) | 16               |
| ix) Overload protection                                                      | 17               |
| x) Endurance (under consideration)                                           | 18               |
| xi) Abnormal operation                                                       | 19               |
| xii) Stability and mechanical hazards                                        | 20               |
| xiii) Mechanical strength                                                    | 21               |
| xiv) Construction                                                            | 22               |
| xv) Internal wiring                                                          | 23               |
| xvi) Components                                                              | 24               |
| xvii) Supply connection and external flexible cables and cords               | 25               |
| xviii) Terminals for external conductors                                     | 26               |
| xix) Provision for earthing                                                  | 27               |
| xx) Screws and connections                                                   | 28               |
| xxi) Creepage distances and clearances                                       | 29               |
| xxii) Resistance to heat, fire and tracking                                  | 30               |
| xxiii) Resistance to rusting                                                 | 31               |
| xxiv) Radiation hazards                                                      | 32               |

**101.2 Acceptance Tests**

The following shall constitute the acceptance tests :

| Tests                                                                     | Clause Reference |
|---------------------------------------------------------------------------|------------------|
| a) Protection against electric shock                                      | 8                |
| b) Input                                                                  | 10               |
| c) Temperature rise                                                       | 11               |
| d) Electrical insulation and leakage current at operating temperature     | 13               |
| e) Moisture resistance                                                    | 15               |
| f) Insulation resistance and electric strength (after humidity treatment) | 16               |
| g) Provision for earthing                                                 | 27               |

NOTE — For the purpose of acceptance tests, the humidity treatment shall be done for 24 hours while conducting the test for moisture resistance (15).

**101.2.1** A recommended sampling procedure for acceptance tests is given in Appendix B of IS 302-1 (1979).

**101.3 Routine Tests**

The following shall constitute the routine tests :

| Tests                                | Clause Reference         |
|--------------------------------------|--------------------------|
| a) Protection against electric shock | 8                        |
| b) High voltage                      | 13.3.2 of IS 302-1(1979) |
| c) Provision for earthing            | 27                       |

**IS 302-2-204 : 1994**

**ANNEX A**  
**TABLES OF TYPE TEST**

This Annex of IS 302-1(1979) is not applicable.

**ANNEX B**  
**SAMPLING PROCEDURE FOR ACCEPTANCE TESTS**

This Annex of IS 302-1(1979) is applicable.

**ANNEX C**  
**ELECTRONIC CIRCUITS**

This Annex of IS 302-1(1979) is applicable if electronic circuits are used.

**ANNEX D**  
**MEASUREMENT OF TEMPERATURE WITH THERMOMETER**

This Annex of IS 302-1(1979) is applicable.

**ANNEX E**  
**ALTERNATIVE TESTS FOR PROTECTED MOTOR UNITS**

This Annex of IS 302-1(1979) is not applicable.

**ANNEX F**  
**IMPACT TEST APPARATUS**

This Annex of IS 302-1(1979) is applicable.

**ANNEX G**  
**THERMAL CONTROLS AND OVERLOAD RELEASES**

This Annex of IS 302-1(1979) is applicable if thermal controls and overload releases are used.

**IS 302-2-204 : 1994**

## **ANNEX H**

### **MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES**

This Annex of IS 302-1(1979) is applicable.

## **ANNEX J**

### **TEST FOR FIRE RESISTING PROPERTIES**

This Annex of IS 302-1(1979) is applicable.

## **ANNEX K**

### **BNF JET TEST FOR DETERMINATION OF THICKNESS OF COPPER AND NICKEL PLATING**

This Annex of IS 302-1(1979) is not applicable.

## **ANNEX L**

### **APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM ON NICKEL, STEEL AND COPPER**

This Annex of IS 302-1(1979) is not applicable.

## **ANNEX AA**

### **LIST OF REFERRED INDIAN STANDARDS**

| <i>IS No.</i> | <i>Title</i>                                                                                                  | <i>IS No.</i>           | <i>Title</i>                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------|
| 302-1 (1979)  | Safety of household and similar electrical appliances : Part 1 General requirements ( <i>fifth revision</i> ) |                         | and for general purposes ( <i>first revision</i> )                                                   |
| 1550 : 1967   | Copper sheet and strip for utensils                                                                           | 1570 (Part 5) :<br>1985 | Schedules for wrought steels : Part 5 Stainless and heat resisting steels ( <i>second revision</i> ) |

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This Indian Standard has been developed from Doc : No. ETD 32 ( 3710 )

### Amendments Issued Since Publication

| Amend No. | Date of Issue | Text Affected |
|-----------|---------------|---------------|
|           |               |               |
|           |               |               |
|           |               |               |
|           |               |               |

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